* Example
Name: Tshilisanani
Age : 28
IS Student : false

23 Prompt Engineering is the skill of writing Clear and Specific instructions to get the best possible responses from AI tools like ChatGPT or Claude.

* Why it's important
- Helps you get accurate and relevant answers
- Saves time by reducing misunderstandings

* Example
• Poor Prompt:
- Tell me about Programming
• Improved Prompt
- Explain the basics of Python Programming for a beginner.

24 MAP functions applies a specific operation to each item in a list and returns a new list with the transformed Values.
- It "Maps" each input to a new output

* Example
numbers = [1, 2, 3, 4
Print # output: [1, 4, 9, 16]

25 Cloud Computing is the delivery of Computing services over the internet instead of using a local Computer.

* Three Main Service Models

1. Iaas
- Provide virtual machines, storage and networks.
- You control the operating system and applications.
* Example: Amazon EC2

2. Paas
- Provides a platform for developing and deploying applications.
- No need to manage hardware or OS
* Example: Google App Engine

3. Saas
- Software delivered over the internet
- Users just use the application
* Example: Gmail

26 A Version Control System is a tool that helps track and manage changes to files over time. It allows ~~the~~ users to see history, revert changes, and collaborat with others.

* Git is a popular distributed version control system used by developers and data analysts to manage code and collaborate on projects efficiently.

* Why Git is widely used
- Tracks changes and history of files
- Enables teamwork without overwriting each other's wo
- Allows branching and merging of code.
- Supports backup and recovery.

* Commit > A saved snapshot of changes in the project at a specific time.
* Branch > A separate version of the project where you can work independently without affecting the main code.

27 A Jupyter Notebook is an open-source tool that allows users to create and share documents containing live code, text, and visualizations.

* Most common used language
- Python.

* Why it is popular among Data Analysts and Data Scientists.
- Combines code, notes, and results in one place.
- Easy to visualize data.
- Supports step-by-step execution of code.

28 Python is a high-level, general-purpose programming language known for simple, readable syntax. It's interpreted, so you can run code line by line without compiling.

* Why it's popular for Data Analytics and AI
- Easy to learn: Clean syntax means less time fighting code, more time on analysis.
- Huge ecosystem: Thousands of libraries for math, data, ML and visualization.

* Four key libraries for data work
1 Pandas : Handles tables and datasets.
Think Excel in code. Great for cleaning, filtering, and merging data.

28 Python
a high level, easy-to-read programming language used for software development, data analysis, and AI

- Simple and easy to learn syntax
- Large community and support
- many powerful libraries for data and AI
- works well with big data and machine learning tools

four python libraries
- ~~AA~~ Numpy
- pandas
- Metlotlib
- Scikit-learn

29. Data privacy
The protection of personal information and how it is collected, stored and used

why it is important
- protects people's personal information
- prevents identity theft and fraud
- Builds trust between users and companies
- ensures data is used fairly and legally

General Data protection
A law that protects personal data and privacy

Give individuals control over their personal
- requires companies to protect and handle data responsible

30. Data Ethics
The branch of ethics that evaluates data practices like collecting, storing, used and shared. It's about doing the right things with data, not just the legal thing.

Unethical examples:
- Biased loan AI: Denies loans to people ~~who~~ from certain areas becaus past data shows higher defaults there.
- Problematics: Punishes individuals for where they live, not their actual creditworthiness.

- Secrete emotion tracking: App uses phone camera to detect users moods without telling them.
- Problematics: Major privacy violation and no consent

- Fake review bots: Company uses AI to flood sites with fake positive reviews.
- Problematics: Misleads customers and destroy trust.

31. Data Governance
- The set of rules, roles and processes an organization uses to manage its data property.

Key Components
- Policies: Rules for data quality, security and access.
- People: Data owners for specific data
- Technology: Tools that enforce the rules and track data.
- Compliance: Making sure data use follows law like POPIA or GDPR

32 Bias in AI
- When an AI system makes unfair decisions that systematically favor or disadvantages certain groups. It happens because the AI learned from flawed or unrepresentative training data.

Biased: The model doesn't treat all people equally. It picked up patterns from human prejudice.

Real-World example: Facial recognition system that struggle to identify people with ~~dak~~ darker skin tones.

Potential Consequences
- Wrongful arrests: Police rely on a misidentification from the AI.
- Denied Services: Users can't unlock phones or access buildings.
- /

33

Role	Main focus	Key Skills
Data Analyst	Looks at past data to find trends and answer business questions.	SQL, Excel, data visualization Statistics basics.
Data Scientist	Predics the future using stats and machine learning models	Python, Machine learning Statistics, Story telling
Data Engineer	Builds and maintains systems that move store data.	SQL, Python, ETL pipeline cloud platforms.
AI Engineer	Takes ML models and puts them into real products / apps.	Software engineering ML deployment APIs, MLOp.

34 Business Intelligence
- The process of turning raw materials / business data into useful insights using software and reports.

Two popular tools
- Power BI: Microsoft tools that connects to Excel, SQL, and other data sources. It makes interactive dashboards and charts. Helps business by letting managers track sales.
- Tableau: Visualization tool known for powerful, drag-and-drop charts. Helps business spot trends fast and share visual reports with teams so decision are based on data, not guesses.

35 The differentces between Data Analyst, Data Scientist, and AI Engineer were most interesting. I thought they were all the same job. Now I understand Data ~~Eage~~ Engineers build the systems, Analysts ~~exan~~ explain the past, and Scientists predict the future. It helped me see which career I might want.

36 Data Governance was hardest forr me at first. All parts like policies, stewards, and compliance sounded very corporate and compliance sounded very corporate and abstract. To understand it better I re-read the notes, watched a ~~you~~ YouTube video called "Data Governance in 5 minutes".

37 I see Data and AI playing a big role in my future career. I want to work as a Data Analyst for a company in South Africa. I'll use tools like SQL and Power BI to help business understand their sales data. I could help a spaza shop in Soweto track which ~~fed~~ products sell best each month so they order the right stock and waste less money.

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4. Artificial intillegence. Machine learning and deep learning

Organization: IBM Cloud learn hub

5. Neural networks and computer vision
Organization: IBM Cloud Learn hub
NVIDIA Developer

6. Natural language Processing
IBM Cloud learn hub

7. APIs and API keys
post man learning center